Appendix A
Hypersegregation

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This appendix outlines the assumptions and segregation measures used in the article *Conceptualizing Topologies of Antiblackness in Education*[1]. In this article, I examine the dimensions of segregation and hypersegregation in the spatial organization of a collection of study regions: $R = \{r_1, \ldots, r_m\}$, where r_k denotes the kth region and and m is the number of regions. Regions are defined using the Metropolitan Statistical Areas (MSAs) reported by the U.S. Census Bureau[2]. Schools are located within regions. We let $S = \{s_1, \ldots, s_q\}$ denote the collection of study schools, where s_ℓ denotes the ℓth school and q is the number of schools. For each region $r_k \in R$, let $S_{r_k} \subseteq S$ denote the schools located within that region.

1 Notation and preliminaries

We let all quantities refer to a specified kth region, $r_k \in R$.

The number of geographic units and all regional totals may differ across regions. A given region contains n mutually exclusive geographic units indexed by i and j. Regional totals are calculated as follows:

$$X = \sum_{i=1}^{n} x_i, \qquad Y = \sum_{i=1}^{n} y_i, \qquad T = \sum_{i=1}^{n} t_i, \qquad A = \sum_{i=1}^{n} a_i.$$

where

x_i is the number of members of the focal group, x, in geographic unit i;

y_i is the number of members of the comparison group, y, in unit i;

t_i is the total population of unit i;

a_i is the land area of unit i;

i and j (for pairwise formulas) index geographic units within region r; and

n is the total number of geographic units in that region;

X and Y are the regional populations of groups x and y, respectively;

T is the total regional population; and

A is the total land area of the region.

Assumption A.1 (Geographic partition). The geographic units are mutually exclusive and collectively exhaust region r. Each member of the population under study is assigned to exactly one unit.

[1] Alexander, Nathan. Forthcoming. Conceptualizing Topologies of Antiblackness in Education.

[2] `https://www.census.gov/data/tables/time-series/demo/popest/2020s-metro-and-micro-statistical-areas-detail`

Assumption A.2 (Group membership and population). Groups x and y are mutually exclusive, so that

$$x_i \geq 0, \qquad y_i \geq 0, \qquad x_i + y_i \leq t_i.$$

If the two groups exhaust the population under study, then $t_i = x_i + y_i$ and $T = X + Y$.

Otherwise, t_i and T include members of additional groups.

Given that we must specify whether our analysis uses the full population or a population restricted to groups x and y, I use a set of simple modifications on variable labels to indicate different specifications.

For example, in the formulation:

$$B = \sum_{i=1}^{n} b_i$$

b_i is used to refer to the number of members of the black population in geographic unit i;

B is the total black population across all units in the region.

$$W = \sum_{i=1}^{n} w_i$$

w_i is used to refer to the number of members of the white population in geographic unit i;

W is the total white population across all units in the region.

When modifications are made to a measure, an explanation is provided as to how variables are defined.

Assumption A.3 (Defined quantities). There are basic rules for some defined quantities.

In all formulas, each denominator is positive wherever used.

I remove empty units such that any unit not meeting some population threshold contributes zero to exposure sums; equivalently, those sums run over units with $t_i > 0$.

Land areas are positive for retained units.

Undefined regional measures are recorded as missing, not assigned zero.

2 Dimensions of segregation

There are five dimensions of segregation defined by Massey and Denton[3] in their 1988 article.

Importantly, some of the dimensions contain multiple measures.

[3]Massey, Douglas S., and Nancy A. Denton. "The dimensions of residential segregation." Social Forces, vol. 67, no. 2, 1988, pp. 281-315. https://doi.org/10.1093/sf/67.2.281

The five dimensions of segregation are: *evenness*, *exposure*, *concentration*, *centralization*, and *clustering*.

Table 1: Dimensions of segregation and measures used in the study

Dimension	Measure	Description
Evenness	Index of Dissimilarity (D)	D compares the proportional distribution of groups across geographic units.
Exposure	Interaction (${}_xP_y$)	${}_xP_y$ measures the average comparison-group, y, share in the unit occupied by a member of x.
	Isolation (${}_xP_x$)	${}_xP_x$ measures the average focal-group, x, share in the unit occupied by a co-member of x.
Concentration	Delta Index (Δ)	Δ compares the focal group's population shares with land-area shares across units.
Centralization	Absolute Centralization Index (ACE_x)	ACE_x measures group x's distribution toward or away from a specified center relative to land area.
Clustering	Spatial Proximity Index (SP)	SP compares population-weighted within-group proximity with overall population proximity. $SP = \frac{XP_{xx} + YP_{yy}}{TP_{tt}}$, where: P_{xx} measures group x's within-group proximity, P_{yy} measures group y's within-group proximity, P_{xy} measure between-group proximity, and P_{tt} is the average proximity among all members of the population under study.

Note: This table lists the measures defined and used in this study, not every available measure within each dimension.

2.1 Evenness

Evenness represents the population dissimilarity between the units in a region r_k. D represents the Index of Dissimilarity, a measure of the uneven distribution of two groups across units within the region and $D \in [0, 1]$.

$$D = \frac{1}{2} \sum_{i=1}^{n} \left| \frac{x_i}{X} - \frac{y_i}{Y} \right| \tag{1}$$

2.2 Exposure

Interaction and isolation measure potential within-unit exposure to group y and group x, respectively:

$${}_xP_y = \sum_{i=1}^{n} \left(\frac{x_i}{X} \right) \left(\frac{y_i}{t_i} \right) \tag{2}$$

$$ {}_xP_x = \sum_{i=1}^{n} \left(\frac{x_i}{X}\right)\left(\frac{x_i}{t_i}\right) \tag{3} $$

You should recall that t_i is the total population in unit i; that is, $t_i \neq T$.

${}_xP_y \in [0,1]$ and ${}_xP_x \in [0,1]$ and depend on regional group composition.

Under uniform composition, ${}_xP_y = \dfrac{Y}{T}$ and ${}_xP_x = \dfrac{X}{T}$.

2.3 Concentration

The Delta Index, Δ, measures the uneven distribution of group x relative to the distribution of land area.

$$ \Delta = \frac{1}{2}\sum_{i=1}^{n} \left|\frac{x_i}{X} - \frac{a_i}{A}\right| \tag{4} $$

2.4 Centralization

The absolute centralization index, ACE_x, measures the distribution of group x around a specified center relative to the distribution of land area. Centralization was commonly measured relative to a central business district (CBD). The Census Bureau uses *distance from the population centroid* rather than the CBD[4].

To retain the original indices for unit i in region r_k, we let π be a permutation of the neighborhood index set $\{1, \ldots, n\}$ that orders neighborhoods by increasing distance from the selected center. Thus, $\pi(h)$ is the original index of the neighborhood in position h. Population and land-area values follow the same ordering.

We first define the cumulative shares as:

$$ C_i^x = \sum_{h=1}^{i} \frac{x_{\pi(h)}}{X}, \qquad C_i^a = \sum_{h=1}^{i} \frac{a_{\pi(h)}}{A}, \qquad C_0^x = C_0^a = 0. $$

where

i is the cumulative endpoint in the ordered sequence;

h indexes positions from 1 through i;

$x_{\pi(h)}$ and $a_{\pi(h)}$ are the focal-group population and land area of the neighborhood occupying position h;

X is the focal group's total regional population;

A is the region's total land area.

[4]U.S. Census Bureau, "Housing Patterns: Appendix B: Measures of Residential Segregation." `https://www.census.gov/topics/housing/housing-patterns/guidance/appendix-b.html`

Thus, C_i^x and C_i^a are the population and land-area shares accumulated through the first i neighborhoods in distance order. The initial shares are zero, and $C_n^x = C_n^a = 1$.

$$ACE_x = \sum_{i=1}^{n} C_{i-1}^x C_i^a - \sum_{i=1}^{n} C_i^x C_{i-1}^a \tag{5}$$

2.5 Clustering

Clustering measures incorporate proximity between units.

We let c_{ij} denote a nonnegative proximity weight between units i and j, with larger values indicating greater proximity. We define average within-group and between-group proximities as follows:

$$P_{xx} = \frac{1}{X^2} \sum_{i=1}^{n} \sum_{j=1}^{n} x_i x_j c_{ij} \tag{6}$$

$$P_{yy} = \frac{1}{Y^2} \sum_{i=1}^{n} \sum_{j=1}^{n} y_i y_j c_{ij} \tag{7}$$

$$P_{xy} = \frac{1}{XY} \sum_{i=1}^{n} \sum_{j=1}^{n} x_i y_j c_{ij} \tag{8}$$

Average proximity among all members of the population under study is

$$P_{tt} = \frac{1}{T^2} \sum_{i=1}^{n} \sum_{j=1}^{n} t_i t_j c_{ij}. \tag{9}$$

For a population consisting of groups x and y, White's Spatial Proximity Index (SP) is

$$SP = \frac{XP_{xx} + YP_{yy}}{TP_{tt}}. \tag{10}$$

3 Hypersegregation

Hypersegregation is defined by Massey and Denton (1989) as a region that exhibits severe segregation in at least four of the five dimensions of segregation[5,6]. Next, the analysis proceeds to examine (a) residential hypersegregation and (b) school hypersegregation among the most populous MSAs.

[5]Massey, Douglas S., and Nancy A. Denton. "Hypersegregation in US Metropolitan Areas: Black and Hispanic Segregation along Five Dimensions." Demography, vol. 26, no. 3, 1989, pp. 373–91.

[6]https://www.ebsco.com/research-starters/sociology/hypersegregation/